Linear Equations - Mark scheme

Algebra · Edexcel IGCSE Mathematics A (4MA1) Paper 1 · Year 10

Question 1 *Practice question written for this worksheet (not from a past paper)*

Solve $7x - 3 = 2x + 12$

$5x = 15$

Answer: x = 3

Question 2 *Practice question written for this worksheet (not from a past paper)*

Solve $4(2x - 1) = 3(x + 6)$

$8x - 4 = 3x + 18$

$5x = 22$

Answer: x = 4.4

Question 3 *Practice question written for this worksheet (not from a past paper)*

Solve $\frac{6}{x-2} = 3$

Multiply by $(x - 2)$: $6 = 3(x - 2)$

$x - 2 = 2$

Answer: x = 4

Question 4 *Edexcel IGCSE Mathematics A (4MA1) · Paper 1HR · June 2019 · Question 4*

4	e.g. $4x - 8x = 17 + 13$ oe		2	M1	For collecting terms in x and number terms on either side of a correct equation
		−7.5		A1	oe e.g. $-\frac{30}{4}$
					Total 2 marks

Question 5 *Practice question written for this worksheet (not from a past paper)*

Solve $\frac{3x-1}{4} - \frac{x+2}{3} = 1$

Multiply by 12: $3(3x-1) - 4(x+2) = 12$

$9x - 3 - 4x - 8 = 12$, so $5x = 23$

Answer: x = 4.6

Question 6 *Practice question written for this worksheet (not from a past paper)*

Sam is 4 years older than Tara. Their ages add up to 30.

Work out Sam's age.

Tara: t; Sam: $t + 4$

$t + t + 4 = 30$, so $t = 13$

Answer: Sam is 17

Question 7 *Practice question written for this worksheet (not from a past paper)*

A bag has x sweets. Anna has 3 bags and 5 extra sweets.

Ben has 2 bags and 17 extra sweets. They have the same number of sweets.

Work out x.

$3x + 5 = 2x + 17$

$x = 12$

Answer: x = 12

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Question 8

Edexcel IGCSE Mathematics A (4MA1) · Paper 4H · June 2016 · Question 2

2	$(8y - 18 =) 3y + 9$ $8y - 3y = 9 + 18$ or $5y = 27$ oe	5.4 oe	3	M1 M1 A1	For correct expansion of bracket For collecting terms in y on one side and constant terms on the other (as part of a correct equation) Eg $\frac{27}{5}$ or $5\frac{2}{5}$ Dep on at least M1 ScB1 for $8y - 18 = 3y + 3$ **AND** $8y - 3y = 3 + 18$ or $8y - 18 = 3y + 3$ **AND** $5y = 21$
	Alternative Method $\frac{8y - 18}{3} = y + 3$ or $\frac{8y}{3} - \frac{18}{3} = y + 3$ oe $\frac{8y}{3} - y = 3 + \frac{18}{3}$ or $5y = 27$ oe	5.4 oe	3	M1 M1 A1	For dividing both sides of the equation by 3 as part of a correct equation For collecting terms in y on one side and constant terms on the other (as part of a correct equation) Eg $\frac{27}{5}$ or $5\frac{2}{5}$ Dep on at least M1
					Total 3 marks

Question 9

Edexcel IGCSE Mathematics A (4MA1) · Paper 4H · January 2014 · Question 3

3.	$18y + 30 = 39$ or $3y + 5 = 6.5$ $18y = 39 - 30$ or $3y = 6.5 - 5$	0.5 oe	3	M1 M1 A1	M1 for correct expansion $\{18y + 30\}$ Dependent on at least one M1
					Total 3 marks

Question 10

Edexcel IGCSE Mathematics A (4MA1) · Paper 4H · January 2012 · Question 3

3.	$4x = 7$ or $4x = 2 + 5$ or $7x - 3x = 7$ oe or $4x - 7 = 0$ oe		3	M2	for correct rearrangement with x terms on one side and numbers on the other AND collection of terms on at least one side or for $4x - 7 = 0$ oe M1 for $7x - 3x = 2 + 5$ oe ie correct rearrangement with x terms on one side and numbers on the other
		$1\frac{3}{4}$ oe		A1	Award full marks for a correct answer if at least 1 method mark scored
					Total 3 marks

Question 11

Edexcel IGCSE Mathematics A (4MA1) · Paper 1F · January 2023 · Question 17

Q	Working	Answer	Mark	Notes
	$6 - 12x$ or $2 - 4x = \frac{5}{3} - \frac{8}{3}x$		3	M1 for expansion of bracket on the LHS or dividing the RHS by 3 with two terms
	$6 - 5 = 12x - 8x$ or $1 = 4x$ or $-12x + 8x = 5 - 6$ oe or $-4x = -1$ or $\frac{8}{3}x - 4x = \frac{5}{3} - 2$ oe or $2 - \frac{5}{3} = -\frac{8}{3}x + 4x$ oe			M1 ft (dep on 4 terms) for terms in x on one side of equation; number terms on the other
	Working required	$\frac{1}{4}$		A1 oe dep on M1 awarded
				Total 3 marks

Question 12

Edexcel IGCSE Mathematics A (4MA1) · Paper 3H · January 2015 · Question 6

6.	$20y - 4 = 18y + 21$		3	M1	for $20y - 4$ **or** $18y + 21$
	$20y - 18y = 21 + 4$ or $2y = 25$			M1	For a correct equation with the y terms collected on one side of the equation and the non y terms on the other side.
		$12\frac{1}{2}$		A1	oe dep on at least M1
					Total 3 marks

Question 13 *Edexcel IGCSE Mathematics A (4MA1) · Paper 4H · June 2013 · Question 6*

6	$5y=14$ **or** $7y-2y=14$ **or** $5y=8+6$ **or** $5y-14=0$	2.8	3	M2	for correct rearrangement with y terms on one side AND correct collection of terms on at least one side or for correct collection to 2 terms
				M1	for correct rearrangement with y terms on one side and numbers on the other eg $7y-2y=8+6$ **OR** correct collection and simplification of either numbers or y terms eg $5y-6=8$ or $5y=a$ or $by=14$
				A1	2.8 oe dependent on at least one M1
					Total 3 marks

Question 14 *Edexcel IGCSE Mathematics A (4MA1) · Paper 3H · January 2016 · Question 8*

8	$6x + 15$			M1	for correct expansion of bracket **OR** division of all terms in a correct equation by 3
	$6x + x = 4 - 15$			M1	for correct rearrangement within a correct equation with x terms on one side and numbers on the other
		$-1\frac{4}{7}$ oe	3	A1	Award full marks for a correct answer if at least 1 method mark awarded (allow $-\frac{11}{7}$ as final answer) accept $-1.57(1428...)$
					Total 3 marks

Question 15 *Edexcel IGCSE Mathematics A (4MA1) · Paper 4H · January 2013 · Question 10*

10.	$5x = -15$ or $5x = 1 - 16$ or $3x + 2x = -15$ or $5x + 15 = 0$		3	M2	for correct rearrangement with x terms on one side and numbers on the other AND correct collection of terms on at least one side M2 also for $-5x = 15$, $-5x = 16 - 1$ or $-2x - 3x = 15$ M1 for correct rearrangement with x terms on one side and numbers on the other eg. $3x + 2x = 1 - 16$ or $16 - 1 = -2x - 3x$ **or** correct collection and simplification of either numbers or x terms eg $.5x + 16 = 1$ or $5x = a$ or $5x - a = 0$ or $nx = -15$ $(n \neq 5)$
		-3		A1	Award 3 marks if M1 scored and answer correct.
					Total 3 marks

Question 16 *Edexcel IGCSE Mathematics A (4MA1) · Paper 2H · January 2023 · Question 13*

Question	Working	Answer	Mark	Notes	
13	eg $20\times\frac{x+3}{4}-20\times\frac{7-x}{5}=20\times4.3$ or eg $5(x+3)-4(7-x)=20\times4.3$ or eg $\frac{5(x+3)}{20}-\frac{4(7-x)}{20}(=4.3)$ or eg $\frac{5(x+3)-4(7-x)}{20}(=4.3)$		3	M1	For clear intention to multiply all terms by 20 (or 4×5) or a multiple of 20 oe or to express LHS as two fractions over 20 (or 4×5) or a multiple of 20 oe or as a single fraction with a denominator of 20 (or 4×5) or a multiple of 20 oe if expanded numerator, allow one error
	eg $5x + 15 - 28 + 4x = 4.3 \times 20$ oe eg $9x - 13 = 86$ eg $9x = 99$			M1	Expanding brackets and multiplying by denominator with no more than one error in total from multiplying out brackets [we must see 4.3×20 or 86 accurately]
	Working required	11		A1	dep on M1
					Total 3 marks

Question 17

Edexcel IGCSE Mathematics A (4MA1) · Paper 3H · January 2013 · Question 2

2. (ii)	$4x + 4 = 12x$ or $2x + 2 = 6x$ $4 = 8x$ or $2 = 4x$	0.5	2	M1 One step from co A1 Allow numerical methods. Correct answer only = M1A1
				Total 4 marks

Question 18

Edexcel IGCSE Mathematics A (4MA1) · Paper 2H · January 2023 · Question 4

Question	Working	Answer	Mark	Notes
4	eg $5x-1=3x+7.4$ oe **or** eg $10x-2+48$ **or** $6x+14.8+48$ **or** $24+24+5x-1+3x+7.4$ oe		4	M1 a correct equation to find x **or** a correct expression for the perimeter in terms of x
	$x = 4.2$			A1 the correct value of x (implies previous mark)
	$2 \times 24 + 2(5 \times$ "4.2" $- 1)$ oe **or** $2 \times 24 + 2(3 \times$ "4.2" $+ 7.4)$ oe **or** $2 \times 24 + (5 \times 4.2 - 1) + (3 \times 4.2 + 7.4)$ oe eg $24 + 24 + 20 + 20$ oe			M1dep on a correct method to find the perimeter – use of positive x from correct working (1st M1 awarded for an equation) **and** only if used the same measurement for AD and BC
	working required	88		A1 cao dep on either M1 or $x = 4.2$
				Total 4 marks

Question 19

Edexcel IGCSE Mathematics A (4MA1) · Paper 4H · January 2015 · Question 8

8	$3x - 5 + 3x + 4x + 2$ $(=10x - 3)$		4	M1 correct expression for perimeter (may be seen in an equation)
	$3x - 5 + 3x + 4x + 2 = 62$ **or** "$10x - 3$"$= 62$			M1 dep
	eg. $10x - 3 = 62$			M1 (dep) correct method to collect x terms **in a correct equation**
		6.5 or $6\frac{1}{2}$		A1 dep on all method marks
				SC : B2 for $x = 6.5$ **and** $3\times6.5 - 5 + 3\times6.5 + 4\times6.5 + 2 = 62$ (B1 for a value for x substituted into correct expression for perimeter eg. $3\times6 - 5 + 3\times6 + 4\times6 + 2$)
				Total 4 marks

Question 20

Edexcel IGCSE Mathematics A (4MA1) · Paper 3H · January 2014 · Question 11

11	Eg. $\frac{4(6x-1)}{4}-\frac{4(5-2x)}{2}=1\times4$ or $6x-1-2(5-2x)=4$ or $\frac{6x-1-2(5-2x)}{4}(=1)$ or $\frac{6x-1}{4}-\frac{2(5-2x)}{4}(=1)$ or $1.5x-0.25-(2.5-x)=1$		4	M1	for clear intention to multiply all terms by 4 or a multiple of 4 or to express LHS as a single fraction with a denominator of 4 or a multiple of 4 or to express LHS as the sum of two fractions with denominators of 4 or a multiple of 4 May be implied by first B1
	Eg. $6x-1-10+4x(=4)$ or $\frac{6x-1-10+4x}{4}(=1)$ or $1.5x-0.25-2.5+x(=1)$			B1	Expanding brackets
	Eg. $10x=15$ or $10x-11=4$ or $10x-1-10=4$ or $6x+4x-11=4$ or $10x-15=0$			B1	for correct rearrangement of a correct equation with terms in x isolated
		1½ oe		A1	Award full marks for a correct answer if at least M1 scored.
					Total 4 marks

Question 21

Edexcel IGCSE Mathematics A (4MA1) · Paper 3H · June 2011 · Question 13

13.	$\frac{20(2x-1)}{4}+\frac{20(x-1)}{5}=2\times20$ or $5(2x-1)+4(x-1)=40$ or $\frac{5(2x-1)+4(x-1)}{20}=2$ or $\frac{5(2x-1)}{20}+\frac{4(x-1)}{20}=2$		4	M1	for clear intention to multiply both sides by 20 or a multiple of 20 or to express LHS as a single fraction with a denominator of 20 or a multiple of 20 or to express LHS as the sum of two fractions with denominators of 20 or a multiple of 20 May be implied by first B1
	$10x-5+4x-4=40$ or $\frac{10x-5+4x-4}{20}=2$ or $\frac{10x-5}{20}+\frac{4x-4}{20}=2$			B1	expanding brackets (dep on M1)
	$14x=49$ or $14x-9=40$ or $10x+4x-9=40$ or $14x-49=0$			B1	dep on both preceding marks ie for a correct rearrangement of a correct equation
		3.5		A1	dep on all preceding marks
					Total 4 marks

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Question 22

Edexcel IGCSE Mathematics A (4MA1) · Paper 4HR · June 2017 · Question 14

Q		Working	Answer	Mark	Notes	
		$\frac{6(5-x)}{2}-\frac{6(x-1)}{3}=6\times 1$ or $3(5-x)-2(x-1)=6$ or $\frac{3(5-x)-2(x-1)}{6}(=1)$ or $\frac{3(5-x)}{6}-\frac{2(x-1)}{6}(=1)$		4	M1	For a clear intention to multiply both sides by 6 or a multiple of 6 or to express LHS as a single fraction with denominator of 6 or a multiple of 6 or to express LHS as the sum of 2 fractions with denominator of 6 or a multiple of 6
		$15-3x-2x+2=6$ or $\frac{15-3x-2x+2}{6}=1$ $\frac{15-3x}{6}-\frac{2x-2}{6}=1$			M1	Expanding brackets in a correct equation.
		E.g. $-3x-2x=6-2-15$ or $-5x=-11$ or $3x+2x=15+2-6$ or $5x=11$			M1	dep on both preceding marks for a correct rearrangement of a correct equation with terms in x on one side and numbers on the other.
			2.2		A1	oe Dependent on M2
						Total 4 marks

Question 23

Edexcel IGCSE Mathematics A (4MA1) · Paper 4H · January 2013 · Question 14

14.	$2(6x+1)=3(5x-2)$ or $\frac{2(6x+1)}{(5x-2)(6x+1)}=\frac{3(5x-2)}{(5x-2)(6x+1)}$		4	M1	Need to see both expressions in an equation May be implied by second M1; NB: Denominators must be correct
	$12x+2=15x-6$ or $\frac{12x+2}{(5x-2)(6x+1)}=\frac{15x-6}{(5x-2)(6x+1)}$			M1	Need to see both expressions in an equation NB: Denominators must be correct
	$3x=8$ or $-3x=-8$ or $3x=2+6$ or $-3x=-6-2$ or $15x-12x=8$ or $12x-15x=-8$ or $3x-8=0$			M1	dep on awarding first two method marks for correct rearrangement with x terms on one side and numbers on the other AND correct collection of terms on at least one side or for $3x-8=0$
		$2\frac{2}{3}$ oe		A1	for $2\frac{2}{3}$ oe including decimal equivalent rounded or truncated to at least 2 decimal places Award 4 marks if first two method marks scored and answer correct.
					Total 4 marks

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CHALLENGE QUESTIONS

Question 24 *Edexcel IGCSE Mathematics A (4MA1) · Paper 3H · June 2012 · Question 9*

9. (b)	Alternative method				
	$\frac{2}{3}y+\frac{1}{3}=\frac{1}{4}y-\frac{1}{2}$		4	M1	for correct expansion
	$\frac{2}{3}y-\frac{1}{4}y=-\frac{1}{2}-\frac{1}{3}$			M1	for correct rearrangement of correct terms
	$\frac{5}{12}y=-\frac{5}{6}$			M1	for correct collection of correct terms on both sides
		−2 oe		A1	Award 4 marks if answer is correct and at least one method mark scored
					Total 7 marks

Question 25 *Edexcel IGCSE Mathematics A (4MA1) · Paper 1FR · June 2024 · Question 10*

Q	Working	Answer	Mark	Notes	
(a)	175 × 1.8 (= 315)		2	M1	for a correct first step
	Correct answer scores full marks (unless from obvious incorrect working)	347		A1	cao
(b)	482 – 32 (= 450)		2	M1	for −32 or ÷ 1.8
	Correct answer scores full marks (unless from obvious incorrect working)	250		A1	cao SCB1 for answer of 464(.2...)
					Total 4 marks

Question 26 *Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · November 2020 · Question 12*

12	**Elimination** E.g. $21x-6y=102$ $21x+35y=-21$ $(-41y=123)$ or $35x-10y=170$ $6x+10y=-6$ $(41x=164)$	**Substitution** E.g. $3\left(\frac{34+2y}{7}\right)+5y=-3$ or $3x+5\left(\frac{7x-34}{2}\right)=-3$ or $7\left(\frac{-3-5y}{3}\right)-2y=34$ or $7x-2\left(\frac{-3-3x}{5}\right)=34$		4	M1	for a correct method to eliminate x or y: coefficients of x or y the same **and** correct operation to eliminate selected variable (condone 1 arithmetical error) **or** for correctly writing x or y in terms of the other variable and correctly substituting
					A1	dep on M1 for $x = 4$ or $y = -3$
	E.g. $7x-2\times-3=34$				M1	dep on M1 for substitution of found variable **or** repeating the steps in first M1 for the second variable
			$x=4$ $y=-3$		A1	cao A correct answer without working scores no marks
						Total 4 marks

Question 27

Edexcel IGCSE Mathematics A (4MA1) · Paper 1HR · June 2024 · Question 12

12	eg $\begin{array}{r} 12x+9y=28.8 \\ 12x+10y=33.6 \end{array}$ or $\begin{array}{r} 20x+15y=48 \\ 18x+15y=50.4 \end{array}$ **or** eg $4x+3\left(\frac{16.8-6x}{5}\right)=9.6$ **or** $6\left(\frac{9.6-3y}{4}\right)+5y=16.8$		4	M1	for multiplication of one or both equation(s) with correct operation selected (allow one arithmetic error) (if + or – is not shown then assume it is the operation that at least 2 of the 3 terms have been calculated for) **or** correct rearrangement of one equation with substitution into the second
		$x=-1.2$ oe **or** $y=4.8$ oe		A1	dep on M1
				M1	(dep on M1) for substitution of found variable **or** repeating the steps in first M1 for the second variable
	Working required	$x=-1.2$ oe **and** $y=4.8$ oe		A1	dep on M1
					Total 4 marks

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Question 28 *Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · November 2021 · Question 6*

Elimination

6	$5a+3p=1.96$ and $3a+2p=1.22$ oe **or** $5a+3p=196$ and $3a+2p=122$ oe		M2 for an arithmetical method (must see the calculation to find 0.22 or 0.26 or 0.74 and 0.48 oe) E.g. 6.1(0) – 5.88 (= 0.22) oe **or** 3.92 – 3.66 (= 0.26) oe **or** 1.96 – 1.22 (= 0.74) oe and 1.22 – "0.74" (= 0.48)	5	M1 for setting up both equations oe Allow the use of apples and pears oe throughout, e.g. 5 apples + 3 pears = 1.96 and 3 apples + 2 pears = 1.22
	E.g. $15a+9p=5.88$ $15a+10p=6.1\ 0$ Subtracting $(-p=-0.22)$	E.g. $10a+6p=3.92$ $9a+6p=3.66$ Subtracting $(a=0.26)$			M1 for a correct method to eliminate a or p: coefficients of a or p the same **and** correct operation to eliminate selected variable (condone any one arithmetic error) **or** to find the cost of 1 apple and 1 pear
	E.g. $5a+3p=1.96$ and $6a+4p=2.44$ oe Subtracting				
	E.g. $5a+3("0.22")=1.96$ or $3a+2("0.22")=1.22$	E.g. $5("0.26")+3p=196$ or $3("0.26")+2p=1.22$	E.g. 3 × 0.22 (= 0.66) 1.96 – "0.66" (= 1.3(0)) "1.3(0)" ÷ 5 (= 0.26) **or** 5 × 0.26 (= 1.3(0)) 1.96 – "1.3(0)" (= 0.66) "0.66" ÷ 3 (= 0.22) **or** Apple and pear is 0.48 oe		M1 (dep on M2) for substituting their value found (must be > 0) of one variable into one of the equations **or** for repeating above method to find second variable **or** for third working column allow $k(a+p)=k(0.48)$ **or** for a complete arithmetical method to find the other value
	E.g $a+p=0.48$ oe				
	$10\times"0.26"+10\times"0.22"$ or $(a+p=)\ 0.48\times10$ oe or $k(a+p)=k(0.48)\times\frac{10}{k}$				M1 (dep on M3) can be implied by $10(a+p)$ provided a and p must be > 0
	Working required			4.8(0)	A1 dep M2
					Total 5 marks

Substitution

6	$5a+3p=1.96$ and $3a+2p=1.22$ oe **or** $5a+3p=196$ and $3a+2p=122$ oe		5	M1 for setting up both equations oe Allow the use of apples and pears oe throughout, e.g. 5 apples + 3 pears = 1.96 and 3 apples + 2 pears = 1.22
	E.g. $3\left(\frac{1.96-3p}{5}\right)+2p=1.22$ or $5\left(\frac{1.22-2p}{3}\right)+3p=1.96$ or $3a+2\left(\frac{1.96-5a}{3}\right)=1.22$ or $5a+3\left(\frac{1.22-3a}{2}\right)=1.96$ or $p = 0.22$ or $a = 0.26$			M1 for correctly writing a or p in terms of the other variable **and** correctly substituting (condone any one arithmetic error)
	E.g. $(a=)\frac{1.96-3(0.22)}{5}$ or $(a=)\frac{1.22-2(0.22)}{3}$ or $(p=)\frac{1.96-5(0.26)}{3}$ or $(p=)\frac{1.22-3(0.26)}{2}$			M1 (dep on M2) for substituting their value found (must be > 0) of one variable into one of the equations **or** for repeating above method to find second variable
	$10\times"0.26"+10\times"0.22"$			M1 (dep on M3) can be implied by $10(a+p)$ provided a and p must be > 0
	Working required	4.8(0)		A1 dep M2
				Total 5 marks